CREDO

Analysing ASTs for Fun and Profit

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neopoly

@rrrene

@rrrene

read in pirate voice

@rrrene

read in pirate voice

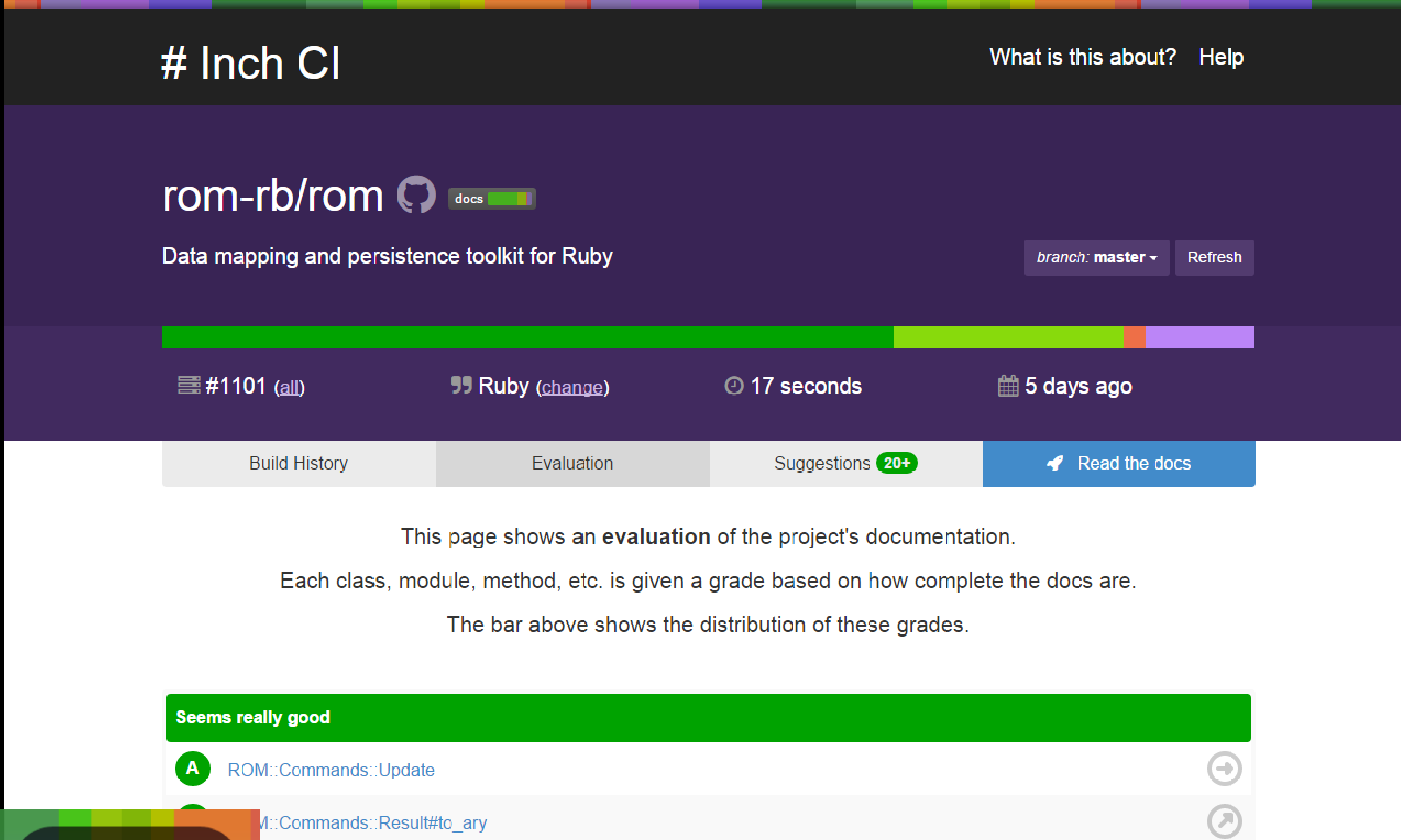
neopoly |> java |> ruby |> elixir

@rrene

read in pirate voice

|> java |> ruby |> elixir

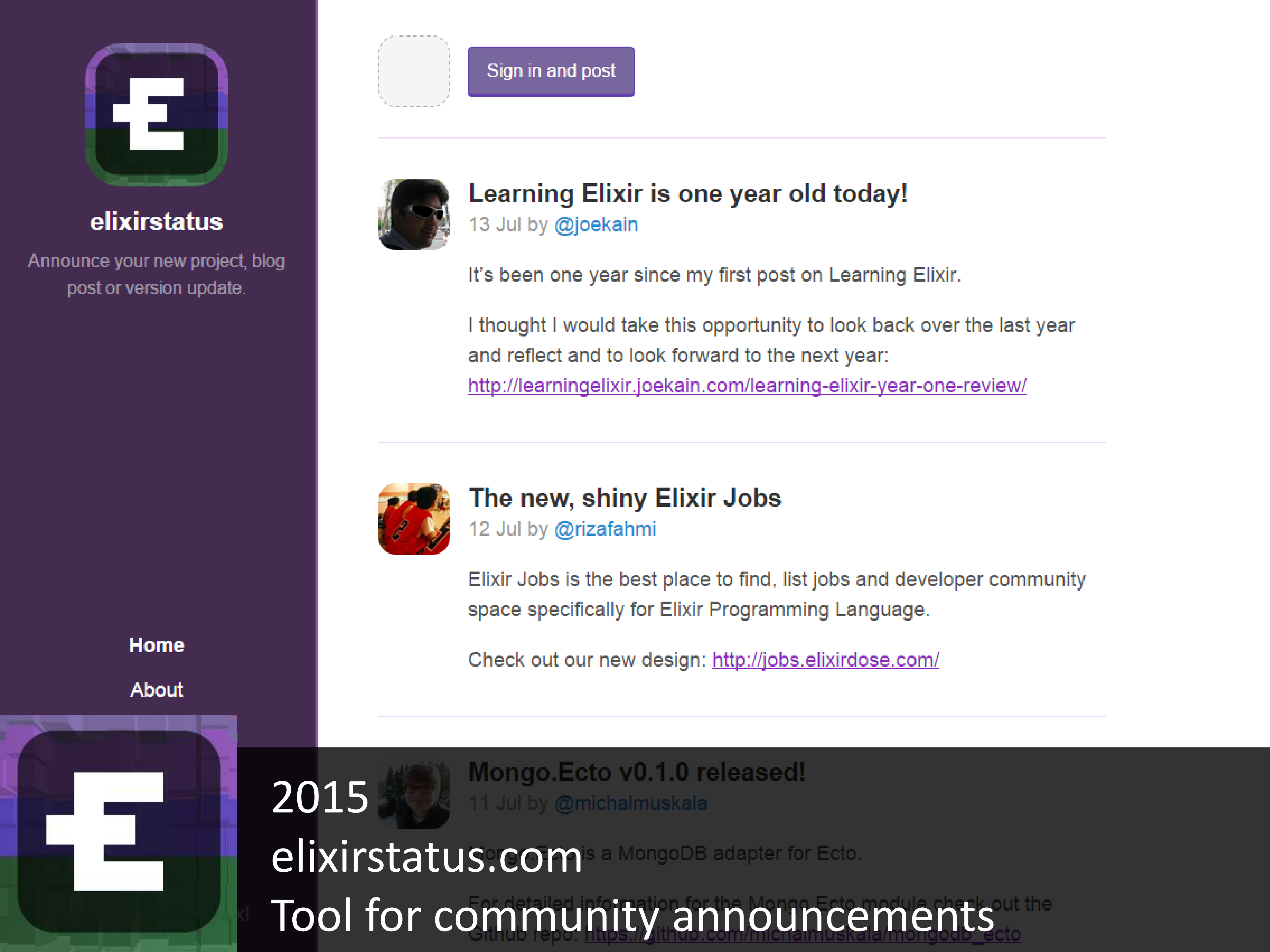
www.neopoly.com/jobs



2014

inch-ci.org

Tool for evaluating inline code documentation



2015
elixirstatus.com
Tool for community announcements

Docs

2014

inch-ci.org

Community

2015

elixirstatus.com

Code

2016

Credo

Today‘s topic!

Excursion:
What is static analysis?

```elixir
def runApiCall(list, mode \\ nil) do
  options = %{}
  list |> IO.inspect
  API.call(list, mode)
end
```

```elixir
11  def runApiCall(list, mode \\ nil) do
12    options = %{}
13    list |> IO.inspect
14    API.call(list, mode)
15  end
```

```
$ mix compile
example.ex:11: warning: default arguments in run/2
are never used
example.ex:12: warning: variable options is unused
```

abstract syntax tree (AST)

{<operation>, <meta>, <arguments>}

abstract syntax tree (AST)

{<operation>, <meta>, <arguments>}

```
# 1 + 2
{:+, [line: 1], [1, 2]}
```

abstract syntax tree (AST)

```
{<operation>, <meta>, <arguments>}
# 1 + 2
{:+, [line: 1], [1, 2]}
# 42 |> inspect()
{:|>, [line: 1], [42, {:inspect, [line: 1], []}]}
```

abstract syntax tree (AST)

```elixir
{<operation>, <meta>, <arguments>}
# 1 + 2
{:+, [line: 1], [1, 2]}
# 42 |> inspect()
{:|>, [line: 1], [42, {:inspect, [line: 1], []}]}
# try it yourself
Code.string_to_quoted!
```

Ruby - RuboCop

Python - pylint

JavaScript - ESLint

Elixir - ?

Ruby - RuboCop

Python - pylint

JavaScript - ESLint

Elixir - Dogma

```
$ mix dogma
Inspecting 78 files.

.X....XXX.XX.XX.XXXXXXXXXXXXX.XXXXXXXXXXXXXXXXXXXXXXX.XXXXXXXXXXXXXXXXXXXXXXXX

78 files, 624 errors!

== lib/plug/adapters/cowboy/conn.ex ==
92: CommentFormat: Comments should start with a single space
82: CommentFormat: Comments should start with a single space
121: FunctionArity: Arity of `parse_multipart` should be less than 4 (was 5).
117: FunctionArity: Arity of `parse_multipart` should be less than 4 (was 5).
38: FunctionArity: Arity of `send_file` should be less than 4 (was 6).
47: LineLength: Line length should not exceed 80 chars (was 92).
49: LineLength: Line length should not exceed 80 chars (was 97).
```

Dogma v0.1.5 analysing Plug v1.2.0-dev (output abbreviated)

(╯°□°)╯︵ ┻━┻

But what is the problem?

Elixir's recent rise from totally unknown to still definitely unknown but mentioned in hushed tones

Clark Kampfe in "Elixir is not Ruby"

https://zeroclarkthirty.com/2015-11-01-elixir-is-not-ruby.html

What‘s the
newcomer experience?

José Valim
@josevalim

Following

Maybe "you are wrong" is enough for talented singers and developers. It is really far from enough for the majority of us. We need guiding.

RETWEETS	LIKES
17	32

3:00 PM - 1 Nov 2015

Let‘s create
a mentoring tool.

```
def runApiCall(list, mode \\ nil) do
  options = %{}
  list |> IO.inspect
  API.call(list, mode)
end
```

breaks convention of underscored names

```
def runApiCall(list, mode \\ nil) do
  options = %{}
  list |> IO.inspect
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end
```

outputs to STDOUT, is slow

breaks convention of underscored names

```elixir
def runApiCall(list, mode \\ nil) do
  options = %{}
  list |> IO.inspect
  API.call(list, mode)
end
```

outputs to STDOUT, is slow

First, we need to categorize issues.

readability

readability

software design

readability
software design
refactoring opportunities

readability
software design
refactoring opportunities
consistency checks

readability
software design
refactoring opportunities
consistency checks
warnings

```elixir
defmodule ReadabilityExample do
  @githubBaseURL "https://github.com/"
  @github_sample_url "https://github.com/rrrene/credo/blob/master/lib/credo/check/consistency/line_endings/unix.ex"
end
```

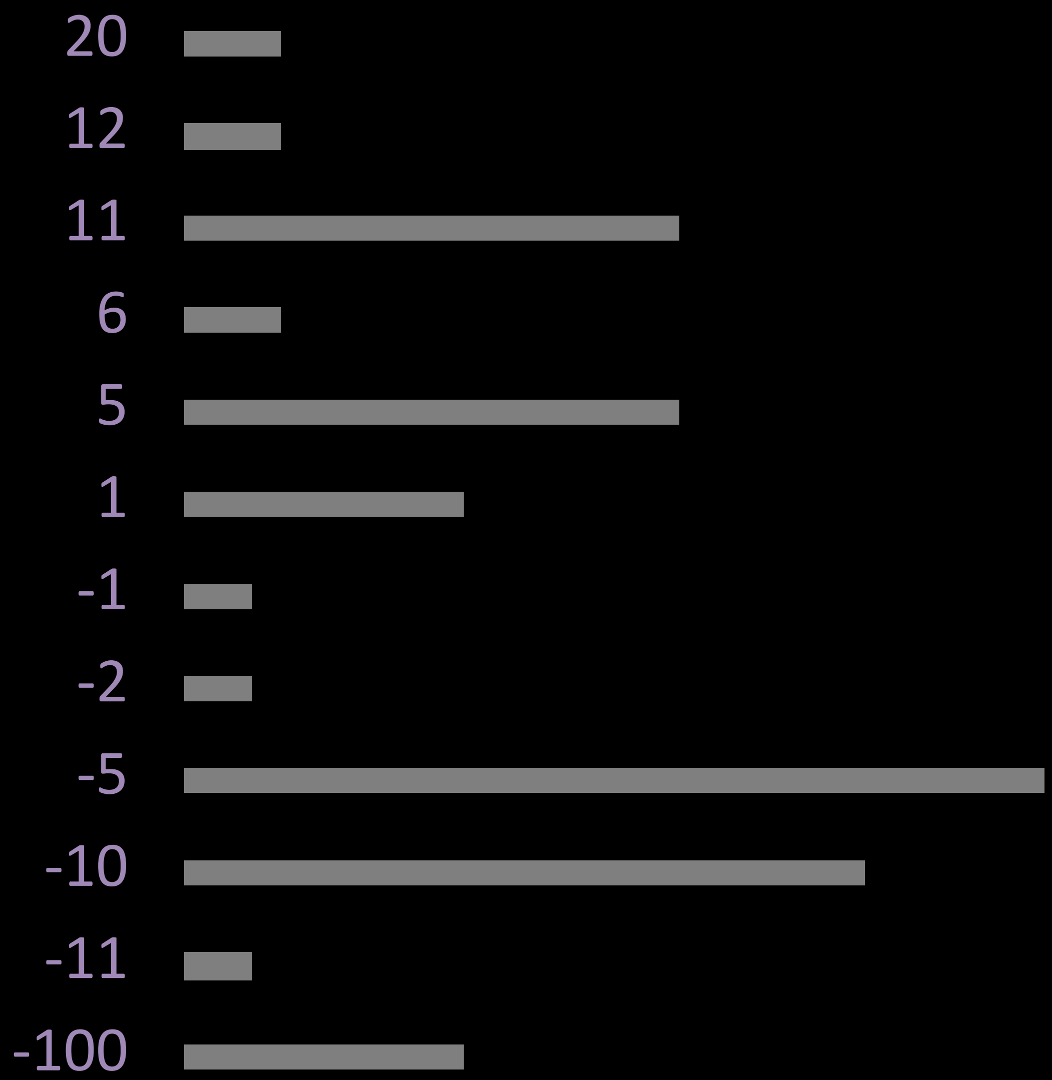
breaks convention of underscored names

```
defmodule ReadabilityExample do
  @githubBaseURL "https://github.com/"
  @github_sample_url "https://github.com/rrrene/
 credo/blob/master/lib/credo/check/consistency/
 line_endings/unix.ex"
end
```

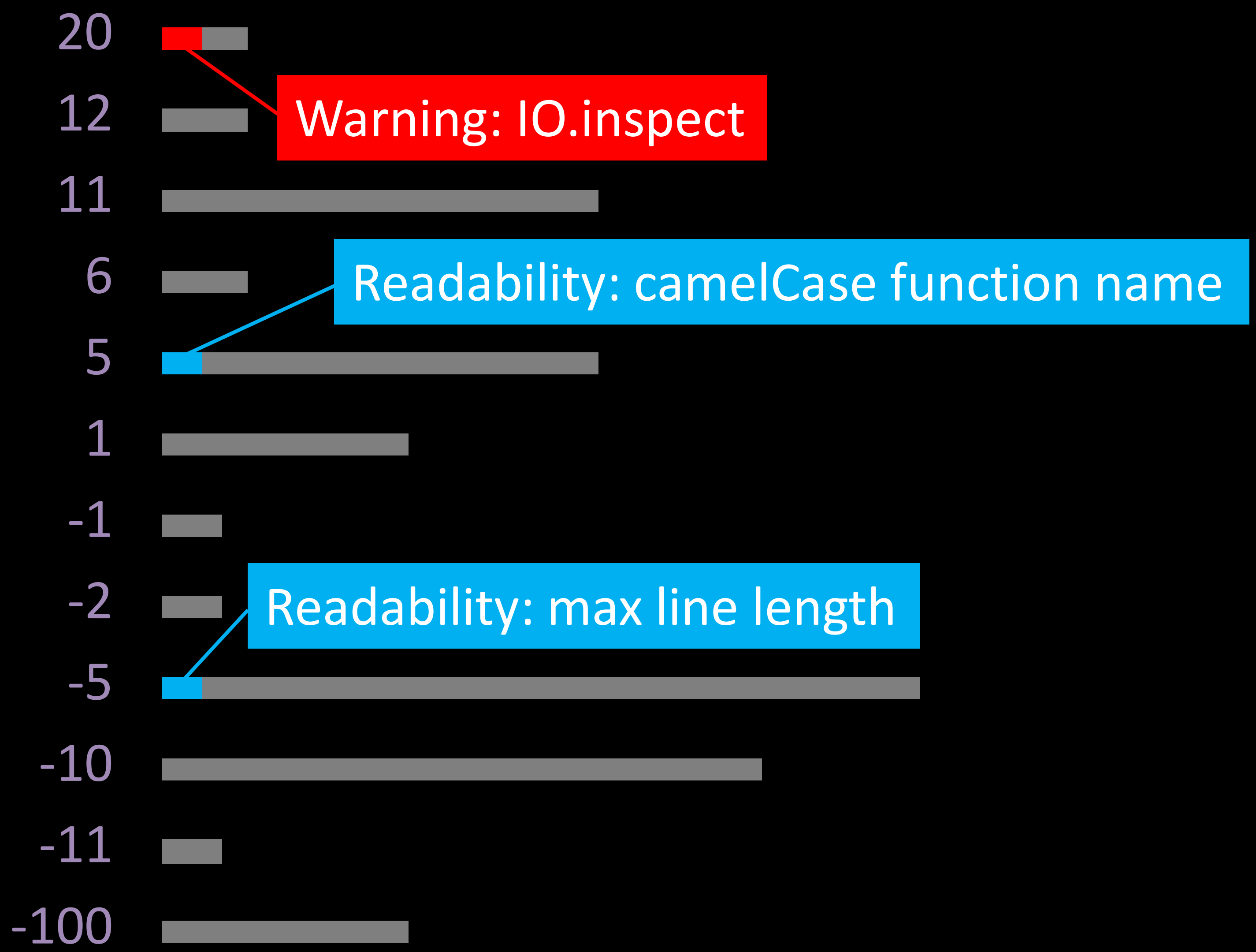
exceeds maximum characters per line

Next, issues need
to be prioritized.

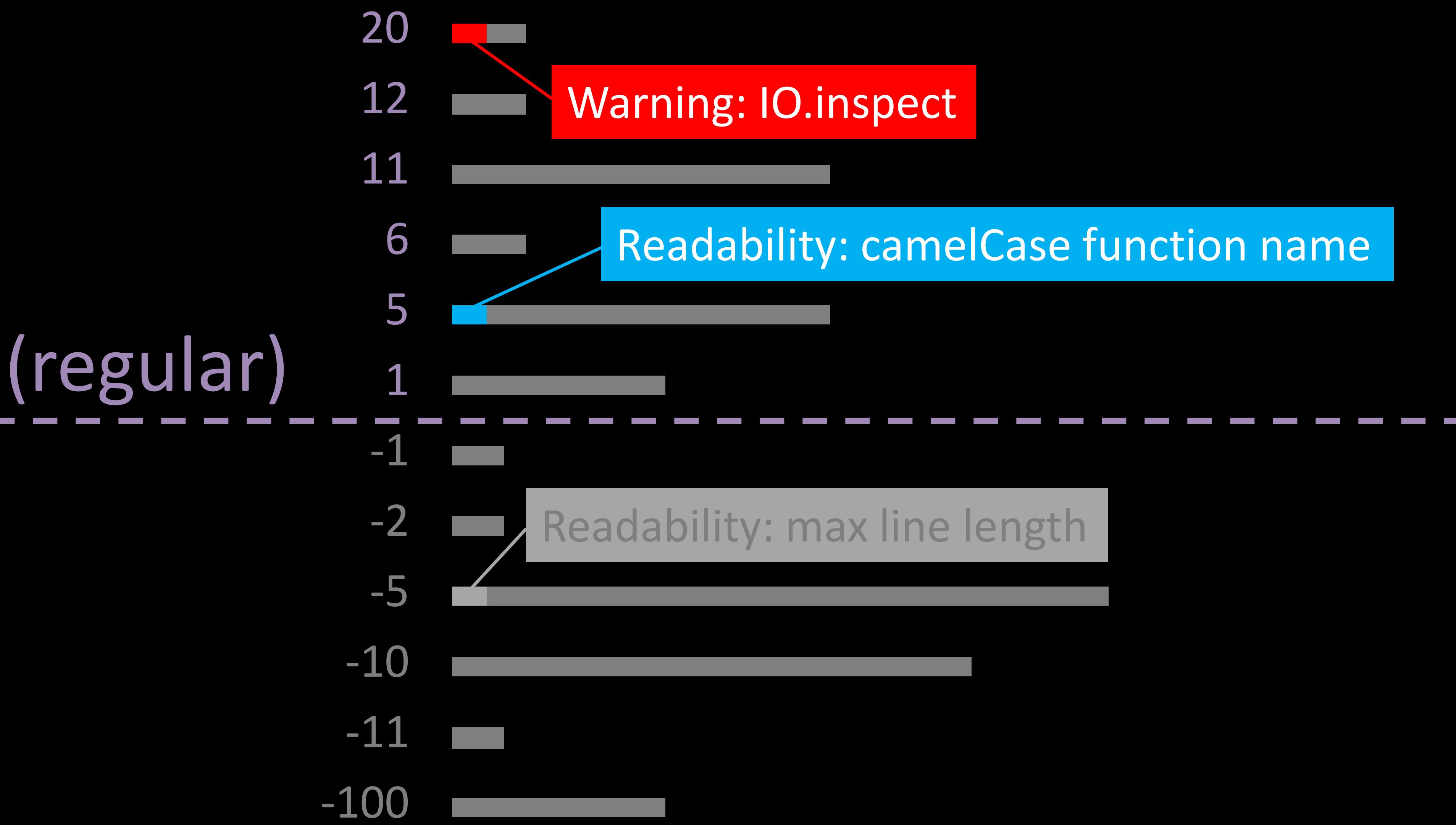
issues ordered by priority



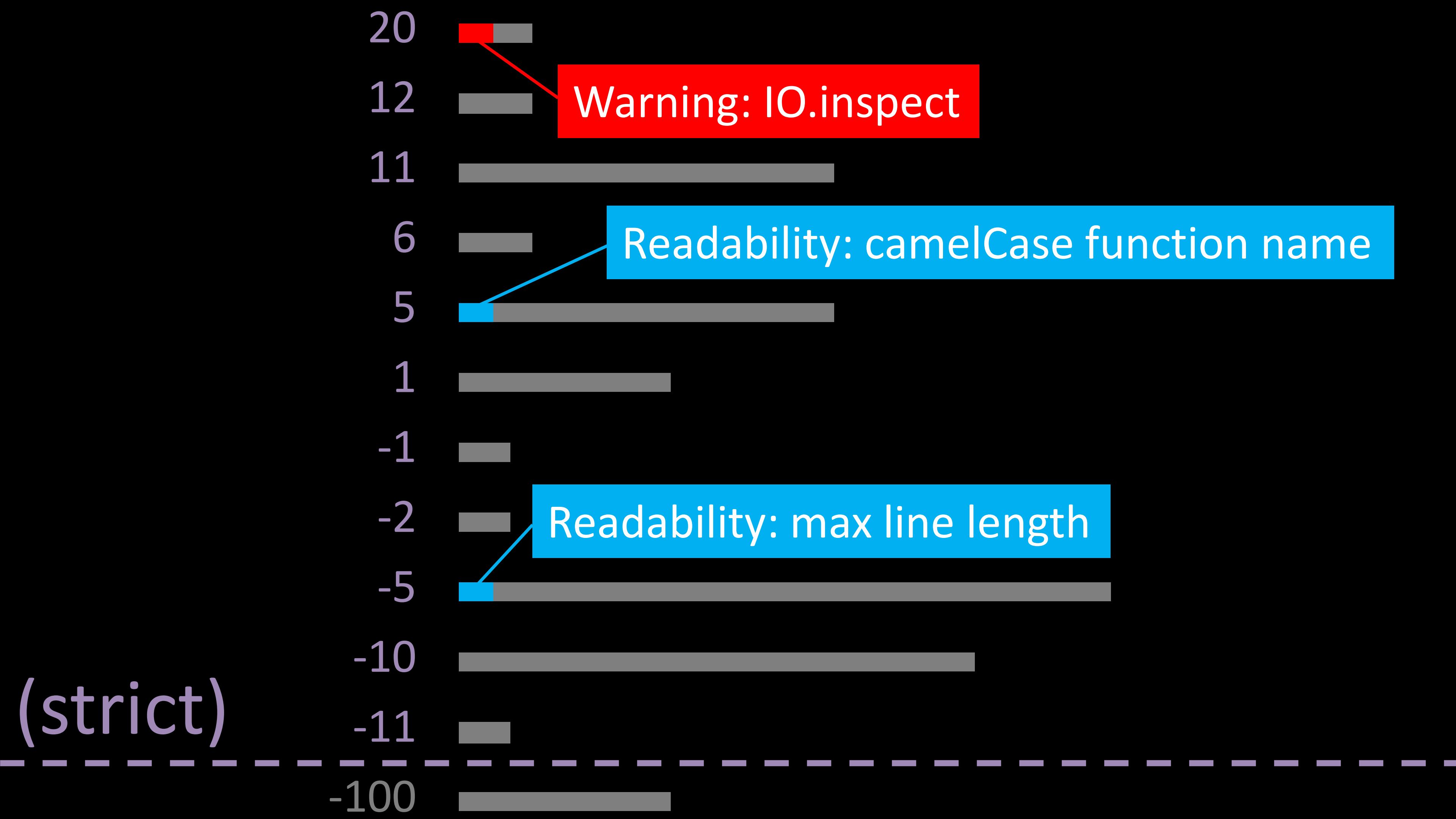
issues ordered by priority



issues ordered by priority



issues ordered by priority



readability

software design

refactoring opportunities

consistency checks

warnings

```
{1, "string", true}
# two ways how to write tuples

# both great, if applied consistently
{ 1, "string", true }
```

without spaces inside the tuple

```
{1, "string", true}
# two ways how to write tuples

# both great, if applied consistently
{ 1, "string", true }
```

without spaces inside the tuple

```
{1, "string", true}
# two ways how to write tuples

# both great, if applied consistently
{ 1, "string", true }
```

with spaces inside the tuple

```elixir
defmodule BadHTTPHeaderError do
  defexception [:message]
end

defmodule UserRequestError do
  defexception [:message]
end
```

```
defmodule BadHTTPHeaderError do
  defexception [:message]
end
```

consistent suffix "Error"

```
defmodule UserRequestError do
  defexception [:message]
end
```

```
defmodule BadHTTPHeaderException do
  defexception [:message]
end

defmodule UserRequestException do
  defexception [:message]
end
```

consistent suffix „Exception"

```
defmodule InvalidHTTPHeader do
  defexception [:message]
end
```

consistent prefix „Invalid"

```
defmodule InvalidUserRequest do
  defexception [:message]
end
```

```
defmodule InvalidHeader do
  defexception [:message]
end
```

no consistent prefix or suffix

fails consistency check

```
defmodule UserRequestFailed do
  defexception [:message]
end
```

```
$ # add credo to mix.exs
$ mix deps.get
$ mix credo
```

```
Checking 44 source files ...

  Code Readability

  [R] → Modules should have a @moduledoc tag.
        lib/plug/upload.ex:1:11 (Plug.UploadError)
  [R] → Modules should have a @moduledoc tag.
        lib/plug/static.ex:110:13 (Plug.Static.InvalidPathError)
  [R] → Modules should have a @moduledoc tag.
        lib/plug/router/utils.ex:1:11 (Plug.Router.InvalidSpecError)

  Refactoring opportunities

  [F] → Function is too complex (ABC size is 49, max is 30).
        lib/plug/adapters/cowboy/conn.ex:94:8 (Plug.Adapters.Cowboy.Conn.parse_multipart)
  [F] → Function takes too many parameters (arity is 7, max is 5).
        lib/plug/static.ex:191:8 (Plug.Static.serve_static)
  [F] → Function takes too many parameters (arity is 6, max is 5).
        lib/plug/parsers.ex:197:8 (Plug.Parsers.reduce)
   ...  (35 more, use `-a` to show them)

  Warnings - please take a look

  [W] ↗ Parameter `path` has same name as a private function in the same module.
        lib/plug/static.ex:173:44 (Plug.Static.serve_static)
  [W] ↗ Parameter `status` has same name as a private function in the same module.
        lib/plug/debugger.ex:142:20 (Plug.Debugger.render)
  [W] ↗ Parameter `length` has same name as the `Kernel.length` function.
        lib/plug/crypto/key_generator.ex:56:74 (Plug.Crypto.KeyGenerator.generate)
   ...  (70 more, use `-a` to show them)

Please report incorrect results: https://github.com/rrrene/credo/issues

Analysis took 0.7 seconds (0.05s to load, 0.7s running checks)
411 mods/funs, found 75 warnings, 40 refactoring opportunities, 3 code readability issues.

Showing priority issues: ↑ ↗ →  (use `--strict` to show all issues, `--help` for options).
```

```
Checking 44 source files ...

  Code Readability

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```

```
$ mix credo lib/plug/static.ex:90:13
```

Plug.Static.InvalidPathError

[R] Category: readability
→ Priority: normal

Modules should have a @moduledoc tag.
lib/plug/static.ex:110:13 (Plug.Static.InvalidPathError)

__ CODE IN QUESTION

```
defmodule InvalidPathError do
          ^^^^^^^^^^^^^^^^
```

__ WHY IT MATTERS

Every module should contain comprehensive documentation.

Many times a sentence or two in plain english, explaining why the module exists, will suffice. Documenting your train of thought this way will help both your co-workers and your future-self.

Other times you will want to elaborate even further and show some examples of how the module's functions can and should be used.

In some cases however, you might not want to document things about a module, e.g. it is part of a private API inside your project. Since Elixir prefers explicitness over implicit behaviour, you should "tag" these modules with

```
@moduledoc false
```

to make it clear that there is no intention in documenting it.

Where do we go from here?

Explanations

Explanations

Style Guide

Explanations

Style Guide

Editor Support

Explanations

Style Guide

Editor Support

Custom Configuration

Explanations

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Custom Configuration

@lint attributes

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Custom Checks are coming!

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Custom Checks are coming!

Creating a Toolbox, not a Linter!

final thoughts

Questions & Alchemists

APPENDIX

Credo – A static code analysis tool with a focus on code consistency and teaching.
https://hex.pm/packages/credo

Dogma – A code style linter for Elixir, powered by shame.
https://hex.pm/packages/dogma

Clark Kampfe in “Elixir is not Ruby”
https://zeroclarkthirty.com/2015-11-01-elixir-is-not-ruby.html

Inch CI – Lint your docs
http://inch-ci.org

ElixirStatus – community announcements
http://elixirstatus.com

HexFaktor – monitor your deps
http://beta.hexfaktor.org